DATABASE TECHNOLOGIES LAB SHEET – 5

Course Code: MCA105P

Topic: Database, Tables and Joins

Q1. Basic Inner Join

Question: Display employee names, department names, and department locations for employees with matching departments.

SQL Code:

```
SELECT Employees.EmpName, Departments.DeptName, Departments.Location
FROM Employees
INNER JOIN Departments
ON Employees.DeptID = Departments.DeptID;
```

Q2. Employee and Department IDs

Question: Use INNER JOIN to display EmpID, EmpName, DeptID, and DeptName.

SQL Code:

```
SELECT Employees.EmpID, Employees.EmpName,
Employees.DeptID, Departments.DeptName
FROM Employees
INNER JOIN Departments
ON Employees.DeptID = Departments.DeptID;
```

Q3. Pharmacy Employees

Question: Use INNER JOIN and WHERE to display employees belonging to the Pharmacy department.

SQL Code:

```
SELECT Employees.EmpName, Departments.DeptName
FROM Employees
INNER JOIN Departments
ON Employees.DeptID = Departments.DeptID
WHERE Departments.DeptName = 'Pharmacy';
```

Q4. Salary Filter

Question: Display matched employees whose salary is greater than 60000.

SQL Code:

```
SELECT Employees.EmpName, Employees.Salary,
Departments.DeptName
FROM Employees
INNER JOIN Departments
ON Employees.DeptID = Departments.DeptID
WHERE Employees.Salary > 60000;
```

Q5. Equi Join

Question: Perform an Equi Join between Employees and Departments using DeptID = DeptID.

SQL Code:

```
SELECT *
FROM Employees
INNER JOIN Departments
ON Employees.DeptID = Departments.DeptID;
```

Q6. Equi Join with Multiple Columns

Question: Display employee name, employee city, department name, and department location using an equality-based join.

SQL Code:

```
SELECT Employees.EmpName, Employees.City,
Departments.DeptName, Departments.Location
FROM Employees
INNER JOIN Departments
ON Employees.DeptID = Departments.DeptID;
```

Q7. Computer Science

Question: Find all employees belonging to Computer Science using an equality-based INNER JOIN.

SQL Code:

```
SELECT Employees.EmpName, Departments.DeptName
FROM Employees
INNER JOIN Departments
ON Employees.DeptID = Departments.DeptID
WHERE Departments.DeptName = 'Computer Science';
```

Q8. Theta Join – Greater Than

Question: Use a Theta Join to display employees whose salary is greater than a salary grade's minimum salary.

SQL Code:

```
SELECT Employees.EmpName, Employees.Salary,
SalaryGrades.Grade, SalaryGrades.MinSalary
FROM Employees
JOIN SalaryGrades
ON Employees.Salary > SalaryGrades.MinSalary;
```

Q9. Theta Join – Salary Range

Question: Assign each employee to the correct salary grade using >= and <= in the join condition.

SQL Code:

```
SELECT Employees.EmpName, Employees.Salary,
SalaryGrades.Grade
```

```
FROM Employees
JOIN SalaryGrades
ON Employees.Salary >= SalaryGrades.MinSalary
AND Employees.Salary <= SalaryGrades.MaxSalary;
```

Q10. Theta Join – Less Than

Question: Display employee names and salary grades where employee salary is less than the grade maximum salary.

SQL Code:

```
SELECT Employees.EmpName, SalaryGrades.Grade,
Employees.Salary, SalaryGrades.MaxSalary
FROM Employees
JOIN SalaryGrades
ON Employees.Salary < SalaryGrades.MaxSalary;
```

Q11. Theta Join – Greater Than or Equal

Question: Display employee name, salary, and grade where employee salary is >= the grade minimum.

SQL Code:

```
SELECT Employees.EmpName, Employees.Salary,
SalaryGrades.Grade
FROM Employees
JOIN SalaryGrades
ON Employees.Salary >= SalaryGrades.MinSalary;
```

Q12. Natural Join

Question: Use NATURAL JOIN to display employee names, DeptID, and department names. Do not write an ON condition.

SQL Code:

```
SELECT EmpName, DeptID, DeptName
FROM Employees
NATURAL JOIN Departments;
```

Q13. Natural Join with Filter

Question: Using NATURAL JOIN, display employees belonging to the department located in Block B.

SQL Code:

```
SELECT EmpName, DeptName, Location
FROM Employees
NATURAL JOIN Departments
WHERE Location = 'Block B';
```

Q14. Natural Join with Salary

Question: Using NATURAL JOIN, display employee name, department name, and salary for salaries >= 70000.

SQL Code:

```
SELECT EmpName, DeptName, Salary
FROM Employees
NATURAL JOIN Departments
WHERE Salary >= 70000;
```

Q15. Inner Join + ORDER BY

Question: Display matched employee name, department, and salary sorted from highest salary to lowest.

SQL Code:

```
SELECT Employees.EmpName, Departments.DeptName,
Employees.Salary
FROM Employees
INNER JOIN Departments
ON Employees.DeptID = Departments.DeptID
ORDER BY Employees.Salary DESC;
```

Q16. DISTINCT Departments

Question: Display distinct department names that have at least one matching employee.

SQL Code:

```
SELECT DISTINCT Departments.DeptName
FROM Employees
INNER JOIN Departments
ON Employees.DeptID = Departments.DeptID;
```

Q17. Employee Count

Question: Use INNER JOIN, GROUP BY, and COUNT to show the number of employees in each matched department.

SQL Code:

```
SELECT Departments.DeptName,
COUNT(Employees.EmpID) AS TotalEmployees
FROM Employees
INNER JOIN Departments
ON Employees.DeptID = Departments.DeptID
GROUP BY Departments.DeptName;
```

Q18. Average Salary

Question: Use INNER JOIN and GROUP BY to calculate average salary for each matched department.

SQL Code:

```
SELECT Departments.DeptName,
AVG(Employees.Salary) AS AverageSalary
FROM Employees
INNER JOIN Departments
ON Employees.DeptID = Departments.DeptID
```

```
GROUP BY Departments.DeptName;
```

Q19. HAVING

Question: Show departments whose average employee salary is greater than 60000.

SQL Code:

```
SELECT Departments.DeptName,
AVG(Employees.Salary) AS AverageSalary
FROM Employees
INNER JOIN Departments
ON Employees.DeptID = Departments.DeptID
GROUP BY Departments.DeptName
HAVING AVG(Employees.Salary) > 60000;
```

Q20. Highest Paid Employee

Question: Find the highest-paid employee among matched employees and display name, department, and salary.

SQL Code:

```
SELECT Employees.EmpName, Departments.DeptName,
Employees.Salary
FROM Employees
INNER JOIN Departments
ON Employees.DeptID = Departments.DeptID
ORDER BY Employees.Salary DESC
LIMIT 1;
```

Q21. IN Condition

Question: Display employees working in either Computer Science or Pharmacy.

SQL Code:

```
SELECT Employees.EmpName, Departments.DeptName
FROM Employees
INNER JOIN Departments
ON Employees.DeptID = Departments.DeptID
WHERE Departments.DeptName IN
('Computer Science', 'Pharmacy');
```

Q22. Multiple WHERE Conditions

Question: Display employees from Delhi or Pune whose salary is at least 50000, with department names.

SQL Code:

```
SELECT Employees.EmpName, Employees.City,
Employees.Salary, Departments.DeptName
FROM Employees
INNER JOIN Departments
ON Employees.DeptID = Departments.DeptID
WHERE Employees.City IN ('Delhi', 'Pune')
AND Employees.Salary >= 50000;
```

Q23. Matched Rows Observation

Question: Use INNER JOIN to display all matched employees. Observe which employee is excluded because no department has DeptID 50.

SQL Code:

```
SELECT Employees.EmpID, Employees.EmpName,
Employees.DeptID, Departments.DeptName
FROM Employees
INNER JOIN Departments
ON Employees.DeptID = Departments.DeptID;
```

Observation: Neha is excluded because her DeptID is 50, and no matching department exists.

Q24. Natural Join + GROUP BY

Question: Using NATURAL JOIN, count employees in each department.

SQL Code:

```
SELECT DeptName, COUNT(EmpID) AS TotalEmployees
FROM Employees
NATURAL JOIN Departments
GROUP BY DeptName;
```

Q25. Final Challenge

Question: Use an equality-based INNER JOIN to find departments whose total employee salary is greater than 100000. Display department name and total salary.

SQL Code:

```
SELECT Departments.DeptName,
SUM(Employees.Salary) AS TotalSalary
FROM Employees
INNER JOIN Departments
ON Employees.DeptID = Departments.DeptID
GROUP BY Departments.DeptName
HAVING SUM(Employees.Salary) > 100000;
```